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TALC: Tail-Aware Label Conversion for Long-Tailed Multi-Label Remote-Sensing Recognition

Jiajun Chen, Jinxin He, Yongbin Yang, Yichun Chen, and Hengshuo Li

***Abstract*—Long-tailed label distributions are common in multi-label remote-sensing recognition, where frequent land-cover labels dominate supervision and rare land-object labels have few positive samples. This imbalance weakens tail-label score support and can remove rare labels from the final image-level label set. Unlike single-label recognition, multi-label prediction must convert class-wise confidence scores into a label set, so a rare label can rank reasonably within its class but still fall below the operating boundary. We propose tail-aware label conversion (TALC), a score-to-label framework for long-tailed multi-label remote-sensing recognition. TALC builds on Asymmetric Loss and adds group-level decision-aware training, using frequency-group SoftF1 and a one-sided tail-recall constraint to align pre-threshold scores with tail-label operating behavior. After model selection, validation-selected operating boundaries are applied to train-only head, medium, and tail groups, forming a compact group-wise rule for label-set generation. On China-MAS-50k with six backbone carriers, TALC-F reaches 0.7200 average Tail F1 under the fixed 0.50 rule, 9.4 percentage points above ASL-F. Its score source attains 0.7925 Tail mAP. Under the same H/M/T group rule, TALC-G reaches 0.7234 Tail F1, 4.7 points above ASL-G. On Agriculture-Vision-2021, TALC-G raises Tail F1 by 5.3 points under the same group rule protocol. These results show that decision-aware score shaping and compact operating boundaries improve tail-label delivery in long-tailed remote-sensing multi-label recognition. Code is available at https://github.com/JiajunChen223/RuralTail-MLR.**

***Index Terms*—Asymmetric Loss, long-tailed learning, multi-label remote-sensing image classification, imbalanced classification**

I. Introduction

HIGH-RESOLUTION remote-sensing images often depict land surfaces composed of multiple land-cover types, land-object categories, and human-made facilities, rather than isolated scene categories. Cropland, roads, water bodies, settlements, greenhouses, and small infrastructure elements may appear within the same image, and their coexistence provides important evidence for image-level interpretation. Scene-level benchmarks and deep remote-sensing surveys have long shown the diversity of visual patterns in aerial and satellite imagery [1], [2]. Multi-label remote-sensing image classification is designed for images where several relevant labels must be retained in one prediction [3], [4]. Together, these labels support monitoring, retrieval, mapping, and regional analysis. Prediction quality in this task depends not only on whether relevant categories receive high confidence scores, but also on whether they remain in the final delivered label set.

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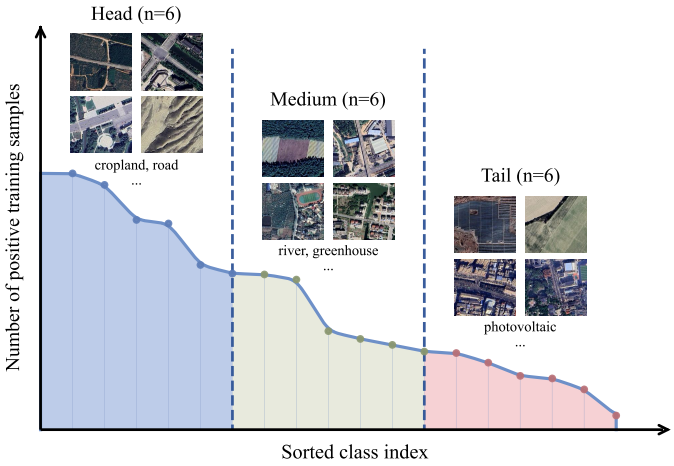


Fig. 1. Training-label frequency distribution on China-MAS-50k. Classes are sorted by training-set occurrence count and partitioned into head, medium, and tail groups using train-only frequency statistics.

This need for complete label delivery is especially pronounced in agricultural and rural observations. Rural scenes contain common background categories, such as cropland, roads, settlements, and water bodies, together with low-frequency objects, such as greenhouses, plastic-covered fields, photovoltaic facilities, sports grounds, dust-proof nets, and environmental objects. These low-frequency categories often appear in fewer images, occupy smaller spatial regions, and vary more strongly in shape and background context. They nevertheless carry practical value for rural construction analysis, agricultural production monitoring, and environmental inspection [5]. When such labels are missed, the image-level label set becomes less useful, even if the dominant background categories are correctly recognized.

Long-tailed label distributions further complicate remote-sensing multi-label recognition. As shown in Fig. 1, China-MAS-50k contains a clear imbalance across categories: a small number of frequent labels provide abundant positive samples, while many tail labels receive limited positive supervision. This imbalance reflects both the natural occurrence of land objects and practical constraints in observation, image collection, and annotation. Under such supervision, standard training is easily dominated by head categories. Tail categories receive weaker score evidence, have fewer opportunities to shape the learned decision function, and often produce lower or less stable confidence scores during inference. More importantly, the final output of multi-label recognition is not the score ranking itself. The score vector must be converted into a discrete label set, and tail labels with useful evidence may still be removed during this conversion.

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Existing long-tailed and imbalanced remote-sensing methods mainly improve rare-category recognition through upstream learning. Data-level strategies alleviate minority scarcity through augmentation or synthetic samples [6], [7]. Representation-learning methods seek features that are less dominated by head categories [8], [9]. Classifier adjustment, reweighting, and decoupled optimization modify the contribution or separation of different classes [10]. Multi-expert frameworks further learn specialized capacities for different frequency regions [11]. These methods help improve score estimates for tail categories and have advanced long-tailed remote-sensing recognition. In multi-label prediction, confidence scores still pass through a decision stage before they become a usable image-level label set. Better score learning alone does not ensure that rare labels will be preserved in the final output.

Under long-tailed distributions, this score-to-label conversion directly affects tail-label delivery. A fixed global operating boundary is stable and widely used, but it implicitly assumes that head and tail categories follow similar score behavior. In practice, head categories usually produce more stable high scores because they have sufficient positive samples, whereas tail categories are learned from sparse evidence and their positive scores are more likely to lie near the operating boundary. A frequency-agnostic operating boundary may remove tail positives that are still recoverable. Fully class-specific operating boundaries provide greater flexibility, but they require reliable validation statistics for every class. When rare categories have very few validation positives, selected operating boundaries can be strongly affected by a small number of validation samples. An overly strict boundary may miss correct tail labels, while an overly loose boundary may introduce false positives and reduce the usability of the predicted label set.

The central issue is the mismatch between score learning and label delivery. Threshold-independent metrics evaluate within-class ranking, but they do not decide which labels enter the final label set. Lowering tail operating boundaries is also insufficient, because correct rare labels and erroneous predictions may increase together. A more effective design should make score learning aware of the decision behavior used after thresholding, while allowing operating boundaries to reflect frequency-dependent score distributions. The decision rule should also avoid unstable per-class searches for labels with very limited validation positives. Group-level operating boundaries address this need by partitioning labels into head, medium, and tail groups according to training-set frequency and selecting one validation-based boundary for each group after model selection. This keeps frequency information within the training protocol while allowing the final decision stage to adapt to different label-frequency regimes.

We propose Tail-Aware Label Conversion (TALC), a score-to-label framework for long-tailed multi-label remote-sensing image classification. TALC uses Asymmetric Loss (ASL) as the base score learner and introduces group-level decision-aware training. Frequency-group SoftF1 encourages pre-threshold scores to reflect group-level precision–recall behavior, while a one-sided tail-recall constraint further protects rare-label evidence during score learning. After model selection, TALC selects operating boundaries on the validation set for head, medium, and tail groups defined only by training-set label frequency. The fixed score model and the selected group-wise boundaries are then applied at test time to generate image-level label sets.

The main contributions of this work are summarized as follows.

1) We study long-tailed multi-label remote-sensing image classification from the perspective of score-to-label conversion, emphasizing that rare-category evidence must be preserved when scores are converted into image-level label sets.
2) We propose TALC, a tail-aware label conversion framework that couples decision-aware score learning with validation-selected H/M/T group-wise operating boundaries under a train-only frequency protocol.
3) We demonstrate consistent improvement in tail-label delivery across six backbone carriers on China-MAS-50k and three representative carriers on Agriculture-Vision-2021, with gains in both tail-label ranking and the delivered image-level label set.

The remainder of this paper proceeds as follows. Section II reviews multi-label remote-sensing recognition and long-tailed learning. Section III formulates the score-to-label problem and train-only frequency grouping. Section IV presents TALC, including its ASL-based score learner, decision-aware objectives, and group-wise operating-boundary selection. Section V reports the multi-carrier and cross-dataset evaluations and the component ablation. Section VI concludes the paper.

II. Related Work

A. Multi-Label Remote-Sensing Recognition

Multi-label remote-sensing research has largely focused on producing reliable class-wise evidence for images containing multiple land elements. Early datasets and retrieval studies moved beyond single scene labels and treated coexisting land categories as image-level annotations [12], [13]. Graph-based retrieval, relation networks, and graph relation learning then modeled label relations more directly [14], [15], [16]. Work on label ambiguity, missing annotations, and noisy supervision further showed that remote-sensing labels can be incomplete or spatially mixed, making label reliability an important part of both training and evaluation [17], [18].

Recent models extend this line by improving how visual evidence is assigned to labels. Concept attention and graph reasoning refine class evidence through semantic relations [19], [20]. Cross-modal label graphs and multidimensional attention strengthen label dependency modeling from complementary feature views [21], [22]. Adaptive graph modules and semantic feature integration further improve class-aware aggregation [23], [24]. Mamba-style sequence modeling and hierarchical crop semantics add efficient sequence modeling and domain-specific label structure [25], [26]. These studies make the score vector more informative before binarization. The subsequent selection of scored labels into the reported label set is usually left to a fixed or separately selected operating rule.

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Recent remote-sensing studies also show how task-aware models exploit spatial, spectral, and cross-domain structure in hyperspectral anomaly detection and object detection [27], [28], [29]. These applications broaden the remote-sensing design landscape, while TALC targets the score-to-label problem in long-tailed multi-label recognition.

B. Long-Tailed and Imbalanced Remote-Sensing Learning

When label frequencies are skewed, score learning is further shaped by head-class dominance. General long-tailed learning has addressed this problem through several recurring mechanisms. Resampling and reweighting alter the contributions of head and tail classes during optimization, while focal-style losses increase attention to difficult examples [30], [31]. Margin adjustment and logit adjustment reshape classifier outputs under imbalanced priors [32], [33]. Decoupled training separates representation learning from classifier learning [34]. Balanced softmax and seesaw losses modify normalization or class contributions to weaken frequent-class bias [35], [36]. Dynamic reweighting follows the same line by adapting class contributions during optimization [37]. In multi-label learning, ASL follows this principle by adapting focal-style modulation to positive-negative imbalance. Distribution-Balanced Loss (DB) further targets long-tailed multi-label recognition through co-occurrence-aware rebalancing and negative-tolerant regularization [38].

Remote-sensing long-tailed studies bring these ideas into aerial and satellite imagery, where rare categories often differ in scale, background, and visual context as well as sample count. The broader imbalance literature in remote sensing includes high-resolution land-cover mapping [39]. In hyperspectral classification, recent work has used class-balancing domain adaptation and sample-generation strategies to address uneven class distributions [40], [41], [42]. Related work on PolSAR and multimodal classification shows that class imbalance also appears across different sensor modalities [43], [44]. Data-side methods enrich the training distribution through augmentation, synthesis, or semantic-aware mixing [45]. Feature-side methods reduce head-class bias in representation learning and transfer more reliable information to tail categories. Classifier-side and framework-level approaches improve class separation or allocate specialized capacity to different frequency regions. EME uses energy-based coarse-grained discrimination and expert specialization to reduce interference across frequency regions in long-tailed remote-sensing recognition [46]. Related studies in semantic segmentation, multimodal object detection, and class-incremental scene classification show the same imbalance pressure across remote-sensing tasks [47], [48], [49].

Taken together, these studies mainly act before score binarization: they add tail evidence, reduce representation bias, rebalance classifiers, or separate expert capacity. In multi-label recognition, the trained learner still produces one score per category. Whether a rare label appears in the delivered label set is then decided by the operating rule applied to that score.

C. Score-to-Label Conversion and Operating Boundaries

Across multi-label recognition and long-tailed learning, the operating rule applied after scoring is often handled as an evaluation choice, not as part of the recognition problem. A fixed global operating boundary provides a clean and reproducible interface for multi-label reporting. A single boundary also imposes the same score requirement on all labels. Under long-tailed training, head labels usually provide more stable positive evidence, while tail labels are more likely to remain near the decision boundary.

Validation-selected operating boundaries add flexibility at this stage. A class-wise boundary can adapt to each label separately, but the estimate becomes fragile when a rare category has only a few validation positives. Small validation-set fluctuations can move the selected boundary and change which tail labels are delivered. Group-wise boundaries provide a more constrained operating rule: labels with similar training frequency share a boundary, reducing the freedom of class-wise search while still separating head, medium, and tail behavior.

This operating stage is where the score vector becomes an image-level label set. For long-tailed multi-label remote-sensing recognition, the boundary is more than a reporting convention; it determines whether weak but useful tail evidence survives into the delivered label set. TALC follows this line by using train-only frequency groups, adding group-level decision-aware score learning, and selecting group-wise boundaries on validation data after model selection.

III. Problem Formulation

A. Score-to-Label Prediction

Let the training set contain N high-resolution images and C non-mutually exclusive labels. For image i and class c, the model outputs a logit z_{ic} and a probability score $p_{ic} = \sigma(z_{ic})$. The delivered prediction is a binarized image-level label set:

$$\hat{y}_{ic}\big(\tau_{g(c)}\big) = \mathbb{I}\left[p_{ic} \geq \tau_{g(c)}\right], \tag{1}$$

where $g(c)$ denotes the frequency group assigned to class c, and $\tau_{g(c)}$ is the score threshold defining its operating boundary. Different choices of $g(c)$ instantiate fixed global, group-wise, or diagnostic class-wise operating rules.

B. Train-Only Frequency Groups

Frequency groups are defined from the training labels. Let f_c be the number of positive training samples for class c. Classes are sorted by f_c and partitioned into three equal-size groups: head G_H, medium G_M, and tail G_T. The group assignment is written as

$$g(c) = \begin{cases} H, & f_c \geq q_{2/3}, \\ M, & q_{1/3} \leq f_c < q_{2/3}, \\ T, & f_c < q_{1/3}. \end{cases} \tag{2}$$

The cutpoints $q_{1/3}$ and $q_{2/3}$ are rank cutpoints from the sorted training frequencies. Group membership depends only on training labels and remains fixed throughout training, validation, and testing.

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C. Evaluation Protocol

We use threshold-free and thresholded metrics to evaluate different stages of multi-label recognition. Threshold-free metrics, including mAP and Tail mAP, are computed from class-wise score rankings before binarization. Thresholded metrics, including Macro-F1 and Tail F1, are computed from the binarized label set after the operating boundaries are applied.

TALC follows this protocol in three steps. First, the score learner is trained with group-level decision signals. Second, the model is selected by validation mAP. Third, H/M/T operating boundaries are selected on the validation set and then frozen for test evaluation. This protocol keeps frequency grouping, model selection, boundary selection, and test reporting separated.

IV. Methodology

A. Overview of TALC

TALC follows the score-to-label formulation in Section III. It does not introduce a new visual backbone; it changes how multi-label scores are trained and converted into the final label set. Fig. 2 summarizes the pipeline. During training, an ASL learner is optimized with two group-level decision-aware terms. During validation, the score model is selected by mAP, and one operating boundary is selected for each train-only H/M/T group. During testing, the backbone, score model, and selected boundaries are frozen.

The train-only head, medium, and tail groups are shared across all stages. They define the groups used for decision-aware training, group-wise boundary selection, and group-wise reporting.

B. ASL Score Learning

ASL is used as the base learner for imbalanced multi-label score learning [50]. It provides the class-wise logits z_{ic} used by the decision-aware module and the subsequent boundary-selection step. TALC keeps the ASL learner unchanged and adds decision-aware terms to its training objective.

All ASL-based score learners use $\gamma_+ = 0$, $\gamma_- = 4$, probability clipping of 0.05, and $\epsilon = 10^{-8}$ under the fixed experimental protocol. The γ_- parameter applies only to ASL-family losses, including ASL and TALC. BCE, Focal Loss, Distribution-Balanced Loss, and EME-ML use their own objectives and do not use this ASL-specific parameter.

C. Decision-Aware Training Module

The decision-aware training module adds two terms to ASL: frequency-group SoftF1 and one-sided tail recall. The reference thresholds define the decision scale used by the differentiable decision-aware objective during training. They are fixed before optimization and remain separate from the final operating boundaries. After validation-mAP model selection, one operating boundary is selected for each train-only H/M/T group on the validation set and frozen before test evaluation.

For each logit z_{ic}, the soft decision is

$$\bar{y}_{ic} = \sigma\left(\frac{z_{ic} - \operatorname{logit}(\bar{\tau}_{g(c)})}{T_s}\right). \tag{3}$$

The temperature T_s controls the smoothness of the transition around the reference threshold.

Given a mini-batch $\mathcal{B}$, the active classes in group g are

$$A_g(\mathcal{B}) = \{c \in G_g : \sum_{i\in\mathcal{B}} y_{ic} > 0\}. \tag{4}$$

For these classes, the soft true-positive, false-positive, and false-negative counts are

$$\begin{cases} \mathrm{STP}_c = \sum_{i\in\mathcal{B}} y_{ic}\bar{y}_{ic}, \\ \mathrm{SFP}_c = \sum_{i\in\mathcal{B}} (1-y_{ic})\bar{y}_{ic}, \\ \mathrm{SFN}_c = \sum_{i\in\mathcal{B}} y_{ic}(1-\bar{y}_{ic}). \end{cases} \tag{5}$$

The class-level SoftF1 score and the group-level average are

$$\begin{cases} s_c^{\mathrm{F1}} = \dfrac{2\mathrm{STP}_c + \epsilon}{2\mathrm{STP}_c + \mathrm{SFP}_c + \mathrm{SFN}_c + \epsilon}, \\ \mathrm{SoftF1}_g = \dfrac{1}{|A_g(\mathcal{B})|} \displaystyle\sum_{c\in A_g(\mathcal{B})} s_c^{\mathrm{F1}}. \end{cases} \tag{6}$$

The mini-batch active group set and its normalizer are

$$\begin{cases} \mathcal{G}_\mathcal{B} = \{g \in \{H, M, T\} : |A_g(\mathcal{B})| > 0\}, \\ Z_\mathcal{B} = \displaystyle\sum_{g\in\mathcal{G}_\mathcal{B}} \alpha_g. \end{cases} \tag{7}$$

The frequency-group SoftF1 loss is

$$L_{\mathrm{gSoftF1}} = \frac{1}{Z_\mathcal{B}} \sum_{g\in\mathcal{G}_\mathcal{B}} \alpha_g(1 - \mathrm{SoftF1}_g). \tag{8}$$

The group weights are fixed as $\alpha_H = 1.0$, $\alpha_M = 1.25$, and $\alpha_T = 1.5$.

The one-sided tail-recall term is defined over the active tail classes:

$$L_{\text{tail-recall}} = \max(0, r_T - \tilde{R}_T)^2, \tag{9}$$

where

$$\tilde{R}_T = \frac{1}{|A_T(\mathcal{B})|} \sum_{c\in A_T(\mathcal{B})} \frac{\mathrm{STP}_c + \epsilon}{\mathrm{STP}_c + \mathrm{SFN}_c + \epsilon}. \tag{10}$$

When $A_T(\mathcal{B})$ is empty, this term is skipped for that mini-batch. The hinge form activates the penalty only when the mini-batch tail recall is below r_T.

The complete training objective is

$$L_{\mathrm{TALC}} = L_{\mathrm{ASL}} + \lambda_{F1} L_{\mathrm{gSoftF1}} + \lambda_R L_{\text{tail-recall}}, \tag{11}$$

where L_{ASL} denotes the standard ASL loss. TALC uses $\lambda_{F1} = 0.2$, $\lambda_R = 0.05$, $T_s = 0.5$, $r_T = 0.65$, $(\alpha_H, \alpha_M, \alpha_T) = (1.0, 1.25, 1.5)$, and reference thresholds $(\bar{\tau}_H, \bar{\tau}_M, \bar{\tau}_T) = (0.50, 0.35, 0.25)$ for all backbone carriers. The reference thresholds are fixed during training and are distinct from the validation-selected operating boundaries used after model selection. The relative weights place group-level SoftF1 as the primary decision-aware term and assign the one-sided tail-recall hinge a smaller tail-directed correction scale.

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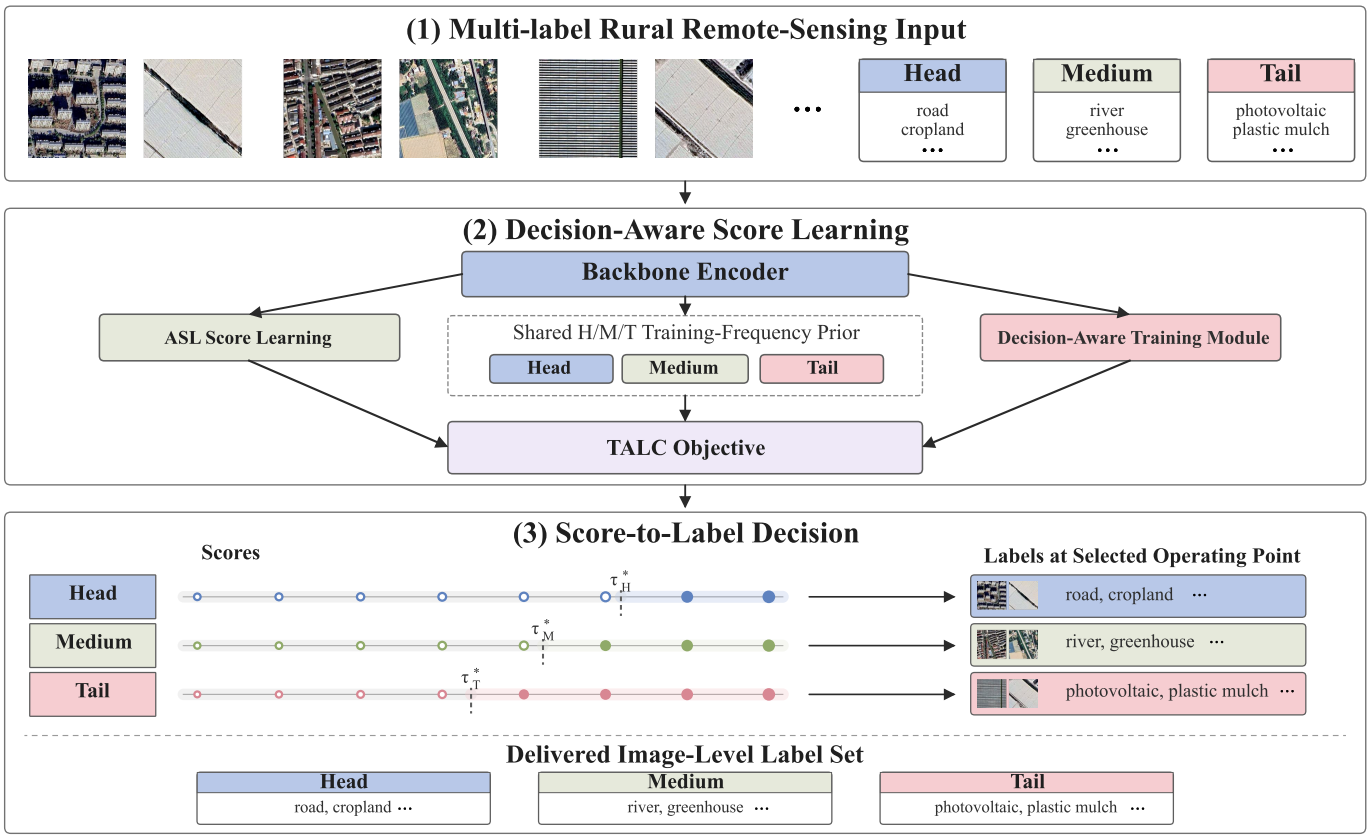


Fig. 2. Overview of the TALC score-to-label pipeline. Multi-label scores are learned with an ASL-based score learner. Train-only frequency groups are reused in decision-aware training and validation-selected group operating boundaries. The final output is a thresholded image-level label set.

D. Group-Wise Operating-Boundary Selection

After model selection, TALC selects operating boundaries on the validation set. The reported group-wise rule uses three boundaries, one for each of G_H, G_M, and G_T. Class-wise operating boundaries are used only for diagnostic analysis.

$$\tau_g^* = \arg\max_{\tau \in \mathcal{T}} \text{Macro-F1}_g^{\text{val}}(\tau), \quad g \in \{H, M, T\}. \tag{12}$$

Here, $\text{Macro-F1}_g^{\text{val}}(\tau)$ denotes the validation Macro-F1 averaged over classes in group g. The search grid is fixed as $\mathcal{T} = \{0.10, 0.15, \ldots, 0.70\}$. The selected boundaries are frozen before test evaluation.

E. Training and Inference Summary

The TALC protocol is fixed before testing. Training optimizes the ASL-based objective with the two decision-aware terms, validation selects the checkpoint by mAP and then the H/M/T boundaries, and test evaluation uses the frozen score model and boundaries. TALC uses no test labels, test-time adaptation, or per-image operating-boundary changes. Algorithm 1 summarizes this sequence.

Algorithm 1 TALC training, validation, and inference.

Require: Training labels; train-only groups G_H, G_M, G_T; reference thresholds $\bar{\tau}_H, \bar{\tau}_M, \bar{\tau}_T$; group weights $\alpha_H, \alpha_M, \alpha_T$; recall target r_T; validation grid $\mathcal{T}$.

Ensure: Final image-level label sets.

1: Compute training-set class frequencies and assign each class to G_H, G_M, or G_T.
2: Initialize the ASL-based score learner.
3: For each training mini-batch $\mathcal{B}$, compute logits and soft decisions using the group reference thresholds.
4: Form $A_g(\mathcal{B})$ for each group, compute weighted SoftF1 for nonempty active groups, and add the tail-recall penalty when active-tail recall is below r_T.
5: Optimize $L_{\text{TALC}} = L_{\text{ASL}} + \lambda_{F1} L_{\text{gSoftF1}} + \lambda_R L_{\text{tail-recall}}$.
6: Select the checkpoint with the highest validation mAP.
7: For each group $g \in \{H, M, T\}$, select $\tau_g^* = \arg\max_{\tau \in \mathcal{T}} \text{Macro-F1}_g^{\text{val}}(\tau)$.
8: Freeze the selected checkpoint and $\{\tau_H^*, \tau_M^*, \tau_T^*\}$.
9: For each inference image, assign each class its train-only group boundary and return $\{c : \mathbb{I}[p_{ic} \geq \tau_{g(c)}^*] = 1\}$.

V. Experiments

A. Datasets and Train-Only Frequency Protocol

China-MAS-50k was used as the primary benchmark for evaluating TALC. It contains 55,520 high-resolution satellite images from 18 rural land-element classes. The dataset covers agricultural fields, rural settlements, transportation elements,

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TABLE I
BACKBONE CARRIERS USED FOR CHINA-MAS-50K EVALUATION.

Carrier	Family	Input	Params	Role
RN50	CNN	384	23.5M	standard baseline
EffV2-S	CNN	384	20.2M	efficient CNN
PVT-B2	Transformer	384	24.9M	hierarchical attention
MOut-S	Mamba-style	384	46.2M	sequence mixing
SFIN-R18	RS-MLR	256	15.2M	spatial relation
MLM-R18	RS-MLR	256	13.0M	label relation

water areas, and several low-frequency rural object classes, making it suitable for evaluating image-level label delivery in rural remote-sensing scenes. We used an 80/10/10 split, with 44,416 images for training, 5,552 for validation, and 5,552 for testing [51].

Agriculture-Vision-2021 was used as the second evaluation dataset [52]. Since the original annotations are semantic masks, we converted them into image-level labels over 9 classes. This conversion creates an image-level evaluation setting based on agricultural aerial imagery and field-level partitioning. A patch was assigned a positive label when the corresponding class mask contained at least one valid positive pixel. The converted 75,278 samples were split by field identifier over the official labeled train/validation pool using an 80/10/10 ratio.

For both datasets, head, medium, and tail groups were formed once from the training split and then kept unchanged across training, validation, and testing. This protocol gives 6/6/6 head, medium, and tail classes on China-MAS-50k and 3/3/3 groups on Agriculture-Vision-2021. Group-aware losses, validation-selected operating boundaries, and group-wise metrics all followed this fixed assignment.

B. Experimental Design and Metrics

The experimental design compares score learners and operating rules under the same data splits, training protocol, and evaluation metrics. On China-MAS-50k, we trained BCE, Focal Loss, ASL, DB, EME-ML, and TALC on six backbone carriers. As shown in Table I, these carriers include convolutional models represented by ResNet and EfficientNetV2 [53], [54], a Transformer-style carrier based on PVTv2 [55], and MOut-S (MambaOut-Small) for testing a recent vision backbone [56]. The table also includes remote-sensing multi-label architectures used as carrier models. This full carrier set was used for the main benchmark evaluation. For Agriculture-Vision-2021, we used RN50, PVT-B2, and MOut-S as representative carriers to test the same protocol on a second dataset.

We implemented EME-ML as a multi-label adaptation of EME [46], using two jointly trained experts for head/medium and tail classes.

Each score learner was evaluated with two operating rules. The fixed rule used an operating boundary of 0.50 for all classes. The group rule selected one operating boundary for each train-only H/M/T group on the validation set, using the grid $\{0.10, 0.15, \ldots, 0.70\}$. In the result tables, the suffix -F denotes the fixed rule, and -G denotes the group rule. Global and class-wise operating boundaries were retained for diagnostic analysis.

All runs used ImageNet-pretrained backbones when available, AdamW optimization, batch size 32, gradient accumulation 2, learning rate 5×10^{-5}, weight decay 0.05, and warmup-cosine scheduling. Training ran for up to 30 epochs with validation-mAP early stopping after a minimum of 12 epochs, patience 6, and $\mathrm{min_delta} = 0.0002$. The checkpoint was selected by validation mAP. Operating boundaries were then selected on the validation set. These settings were shared across all compared learners.

We reported mAP and Tail mAP before thresholding, and Macro-F1 and Tail F1 after thresholding. The mAP metrics evaluate score ranking, while the F1 metrics evaluate the image-level label set produced by each operating rule.

C. Main Results on China-MAS-50k

Table II reports the carrier-level comparison of BCE, Focal Loss, ASL, DB, EME-ML, and TALC. TALC achieves the highest Tail F1 under both the fixed and H/M/T group rules, while its learned score source achieves the highest Tail mAP among the compared learners.

The ordering is consistent across convolutional, Transformer-style, Mamba-style, and remote-sensing-specific carriers, so the gain is not tied to a single backbone family.

Table III reports the class-wise shift from ASL-G to TALC-G under the same group rule protocol. Head-class performance remains nearly unchanged, while the largest average gains appear in the tail block. This localizes the main improvement to tail labels rather than to a broad change in all classes. The class-level rows show different recovery patterns: dust-proof net improves mainly through recall, plastic mulch gains both precision and recall, and photovoltaic achieves a higher F1 despite a 0.2-point AP decrease.

Table IV summarizes six-carrier fixed-to-group shifts and carrier-averaged H/M/T boundaries selected from the fixed grid $\mathcal{T} = \{0.10, 0.15, \ldots, 0.70\}$. BCE and Focal Loss recover more tail positives after group rule selection, but lose more precision. ASL also improves under the group rule, with a larger movement from its fixed operating point. TALC starts from the highest fixed 0.50 tail result and changes only modestly after group boundary selection.

D. Score-Source and Operating Analysis

The China-MAS-50k results can be separated into two components: the score source learned by the model and the operating rule used to form the final label set. Fig. 3 follows the fixed-to-group pattern in Table IV. Panel (a) shows the score-to-label gap under fixed 0.50 decisions, panel (b) keeps ASL scores fixed and changes the operating rule, while panels (c) and (d) replace the score source with TALC scores and then apply the selected H/M/T boundaries. Table V reports the corresponding metric changes relative to ASL-F.

ASL-G isolates the operating-rule effect: the ASL score source is fixed, ranking metrics stay unchanged, and Tail F1 increases by 5.1 percentage points. TALC-F isolates the score-source effect: the fixed 0.50 rule is unchanged, while overall mAP, Tail mAP, and Tail F1 increase together. TALC-G then applies the selected H/M/T boundaries to the TALC score

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TABLE II
CARRIER-RESOLVED TAIL F1 AND TAIL MAP UNDER FIXED AND GROUP RULES ON CHINA-MAS-50K.

Learner	MLM-R18			SFIN-R18			RN50			MOut-S			EffV2-S			PVT-B2			Avg.		
	F1-F	F1-G	mAP	F1-F	F1-G	mAP	F1-F	F1-G	mAP	F1-F	F1-G	mAP	F1-F	F1-G	mAP	F1-F	F1-G	mAP	F1-F	F1-G	mAP
BCE	0.5821	0.5928	0.6157	0.6373	0.6642	0.7259	0.6084	0.6134	0.6609	0.6996	0.7104	0.7878	0.6883	0.7135	0.7742	0.7014	0.7571	0.8180	0.6529	0.6752	0.7304
Focal	0.5858	0.6308	0.6439	0.5208	0.6133	0.6939	0.6332	0.6790	0.7520	0.6301	0.7337	0.8051	0.5946	0.7051	0.7642	0.6242	0.7486	0.8111	0.5981	0.6851	0.7450
ASL	0.5347	0.5758	0.6009	0.5538	0.6125	0.7002	0.6285	0.6964	0.7640	0.6481	0.7125	0.7933	0.6750	0.7323	0.8010	0.7171	0.7318	0.8085	0.6262	0.6769	0.7447
DB	0.6291	0.6341	0.6556	0.6566	0.6566	0.7101	0.7078	0.6922	0.7881	0.7375	0.7375	0.8062	0.7260	0.7260	0.7944	0.7194	0.7420	0.8196	0.6961	0.6981	0.7623
EME-ML	0.5989	0.6104	0.6447	0.6444	0.6641	0.7030	0.6993	0.6993	0.7594	0.7251	0.7329	0.7889	0.6965	0.6965	0.7720	0.7247	0.7343	0.7977	0.6815	0.6896	0.7443
TALC	**0.6591**	**0.6665**	**0.7206**	**0.6841**	**0.6868**	**0.7685**	**0.7271**	**0.7271**	**0.7959**	**0.7639**	**0.7664**	**0.8223**	**0.7359**	**0.7439**	**0.8172**	**0.7496**	**0.7499**	**0.8306**	**0.7200**	**0.7234**	**0.7925**

TABLE III
CLASS-WISE SHIFTS FROM ASL-G TO TALC-G ON CHINA-MAS-50K. DELTA VALUES ARE PERCENTAGE POINTS.

Head						Medium						Tail					
Class	Pos.	ΔAP	ΔP	ΔR	ΔF1	Class	Pos.	ΔAP	ΔP	ΔR	ΔF1	Class	Pos.	ΔAP	ΔP	ΔR	ΔF1
Cropland	2996	+0.0	+0.2	+0.0	+0.2	Grassland	149	+2.1	-8.4	+12.4	+3.2	Solid waste	97	+6.8	-4.2	+9.3	+2.8
Road	1496	-0.2	+1.5	-2.1	-0.2	Factory bldg.	598	+0.1	+2.3	-2.1	+0.1	Park	122	+0.3	+6.0	-3.1	+2.1
Lake/pond	608	-0.0	+1.4	-1.5	-0.4	Greenhouse	188	+0.5	+5.9	-3.5	+1.7	Basketball court	145	+7.8	+3.9	+8.2	+6.5
Woodland	3683	+0.0	-0.9	+0.1	-0.4	River	545	+0.7	+4.5	-3.9	+0.2	Dust-proof net	47	+11.6	-1.6	+18.1	+11.9
Bare land	709	+0.5	+1.1	+0.5	+1.3	Railway	216	+1.1	+8.1	-1.9	+3.6	Plastic mulch	92	+2.5	+5.5	+3.3	+4.3
Rural village	1595	+0.2	-1.2	+1.3	+0.0	Football field	168	+1.2	-1.1	+1.9	+0.4	Photovoltaic	75	-0.2	+1.8	-0.9	+0.3
Avg.	–	+0.1	+0.3	-0.3	+0.1	Avg.	–	+0.9	+1.9	+0.5	+1.5	Avg.	–	+4.8	+1.9	+5.8	+4.7

TABLE IV
CARRIER-AVERAGED FIXED-TO-GROUP TAIL OPERATING SHIFTS AND H/M/T BOUNDARIES ON CHINA-MAS-50K.

Learner	Tail mAP	Fixed boundary				Carrier-averaged H/M/T boundaries				Shift		
		τ	P	R	F1	H/M/T τ	P	R	F1	ΔP (pp)	ΔR (pp)	ΔF1 (pp)
BCE	0.7304	.50	0.8254	0.5663	0.6529	.32/.28/.32	0.7478	0.6334	0.6752	-7.8	+6.7	+2.2
Focal	0.7450	.50	**0.8951**	0.4889	0.5981	.35/.36/.33	0.7303	0.6672	0.6851	-16.5	+17.8	+8.7
ASL	0.7447	.50	0.5401	**0.7853**	0.6262	.64/.64/.64	0.7320	0.6553	0.6769	+19.2	-13.0	+5.1
DB	0.7623	.50	0.7691	0.6540	0.6961	.49/.48/.49	0.7684	0.6558	0.6981	-0.1	+0.2	+0.2
EME-ML	0.7443	.50	0.7920	0.6208	0.6815	.50/.48/.37	0.7660	0.6459	0.6896	-2.6	+2.5	+0.8
TALC	**0.7925**	.50	0.7454	0.7152	0.7200	.62/.53/.53	0.7510	0.7132	**0.7234**	+0.6	-0.2	+0.3

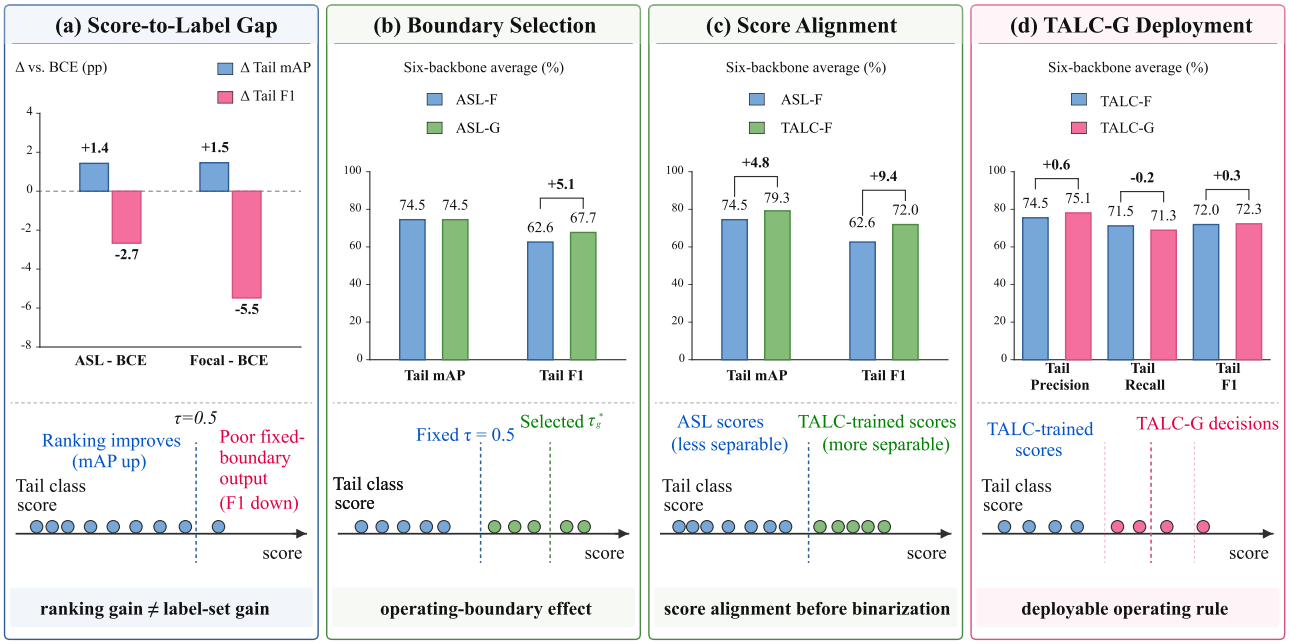
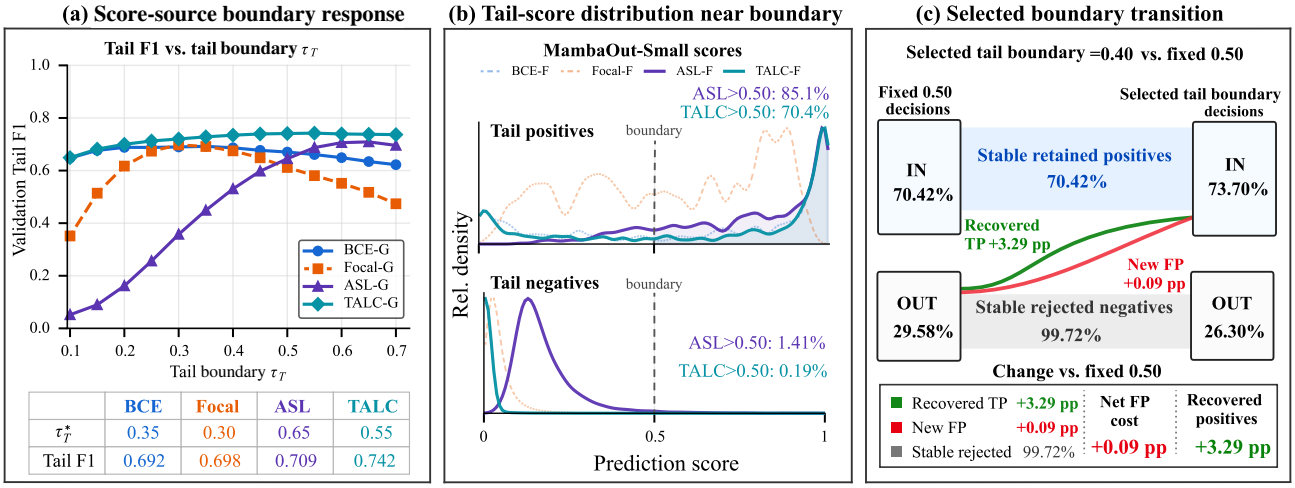


Fig. 3. Score source and operating rule decomposition on China-MAS-50k. (a) Score-to-label gap for ASL and Focal relative to BCE under the fixed 0.50 rule. (b) Boundary selection from ASL-F to ASL-G. (c) Score alignment from ASL-F to TALC-F under the fixed 0.50 rule. (d) TALC-G deployment after applying validation-selected H/M/T operating boundaries to TALC-trained scores. Bars show six-backbone averages, and the lower sketches summarize the corresponding score-to-label rules.

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	BCE	Focal	ASL	TALC
$\tau_T^\star$	0.35	0.30	0.65	0.55
Tail F1	0.692	0.698	0.709	0.742

Fig. 4. Score-source response and operating-boundary behavior on China-MAS-50k. (a) Validation Tail F1 over tail operating boundaries for BCE-G, Focal-G, ASL-G, and TALC-G. (b) Gaussian KDEs of Tail-positive and Tail-negative confidence values near the global 0.50 boundary for BCE-F, Focal-F, ASL-F, and TALC-F on MambaOut-Small. The dashed and solid vertical lines mark the global and validation-selected tail boundaries, respectively. (c) Transition from the fixed 0.50 rule to the selected tail boundary for a representative MambaOut-Small TALC-G case.

TABLE V
SCORE SOURCE, OPERATING RULE, AND DIAGNOSTIC OPERATING-BOUNDARY ANALYSIS ON CHINA-MAS-50K.

Variant	Overall				Tail						H/M/T τ
	mAP	ΔmAP	Macro-F1	ΔMacro-F1	mAP	ΔmAP	P	R	F1	ΔF1	
ASL-F	0.8125	+0.000	0.7063	+0.000	0.7447	+0.000	0.5401	**0.7853**	0.6262	+0.000	.50/.50/.50
ASL-G	0.8125	+0.000	0.7450	+0.039	0.7447	+0.000	0.7320	0.6553	0.6769	+0.051	.64/.64/.64
TALC-F	0.8320	+0.020	0.7614	+0.055	0.7925	+0.048	0.7454	0.7152	0.7200	+0.094	.50/.50/.50
TALC-G	**0.8320**	+0.020	0.7659	+0.060	**0.7925**	+0.048	0.7510	0.7132	0.7234	+0.097	.62/.53/.53
TALC-Global	0.8320	+0.020	0.7632	+0.057	0.7925	+0.048	0.7635	0.6990	0.7200	+0.094	.58/.58/.58
TALC-Class-wise	0.8320	+0.020	**0.7681**	+0.062	0.7925	+0.048	**0.7837**	0.6902	**0.7290**	+0.103	–

source and reports the compact group-wise score-to-label rule used for final evaluation.

The component ablation separates the two decision-aware terms under the same ASL score learner and training protocol. The validation results are reported as mean ± standard deviation over three random seeds, with Macro-F1 and Tail F1 computed using the validation-selected H/M/T boundaries of each seed and the test split reserved for the main comparison. Frequency-group SoftF1 raises the mean Tail F1 from 0.7185 for ASL to 0.7355 for SoftF1-only.

Full TALC reaches 0.7360 with both terms enabled, whereas TailRecall-only reaches 0.7181. SoftF1 drives the gain, while tail recall supplies the explicit one-sided tail constraint in the full formulation.

TABLE VI
COMPONENT ABLATION ON THE CHINA-MAS-50K VALIDATION SET.

Variant	mAP	Macro-F1	Tail F1
ASL	0.8204 ± 0.0039	0.7557 ± 0.0025	0.7185 ± 0.0061
SoftF1-only	0.8282 ± 0.0040	0.7652 ± 0.0063	0.7355 ± 0.0146
TailRecall-only	0.8197 ± 0.0029	0.7552 ± 0.0020	0.7181 ± 0.0076
Full TALC	$\mathbf{0.8285 \pm 0.0028}$	$\mathbf{0.7656 \pm 0.0052}$	$\mathbf{0.7360 \pm 0.0170}$

Fig. 4 examines operating-boundary behavior in three views. Panel (a) plots the six-carrier mean validation Tail F1 across tail operating boundaries for each score source. Panel (b) compares Gaussian KDEs of Tail-positive and Tail-negative scores near the fixed 0.50 boundary. Panel (c) shows the transition from the fixed boundary to the selected TALC-G tail boundary for a representative MambaOut-Small case.

Averaged over the six carriers, the validation Tail F1 response peaks near 0.35 for BCE-G, 0.30 for Focal-G, 0.65 for ASL-G, and 0.55 for TALC-G. In the representative MambaOut-Small test distribution, TALC-F reduces tail-negative spillover above 0.50 from 1.41% for ASL-F to 0.19%, while retaining 70.4% of tail positives. Moving from the fixed 0.50 rule to the selected TALC-G tail boundary of 0.40 increases retained tail positives from 70.42% to 73.70%, with an additional false-positive cost of 0.09 percentage points.

E. Second-Dataset Evaluation on Agriculture-Vision-2021

Agriculture-Vision-2021 extends the evaluation to high-resolution agricultural aerial imagery with a different label construction from China-MAS-50k. The same train-only H/M/T grouping protocol and validation-selected group rule were used.

Across RN50, PVT-B2, and MOut-S, TALC-G achieves the highest Tail F1 in Table VII. Averaged over the three carriers, it raises Tail F1 from 0.6008 for ASL-G to 0.6541 and Tail mAP from 0.6535 to 0.7387. The Tail F1 gain is 5.3 percentage points under the same group rule protocol. TALC-G also achieves the highest average mAP and Macro-F1 among the compared variants.

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TABLE VII
SECOND-DATASET EVALUATION ON AGRICULTURE-VISION-2021.

Method	RN50	PVT-B2	MOut-S	Avg.						
	Tail F1	Tail F1	Tail F1	mAP	Macro-F1	Tail P	Tail R	Tail F1	Tail mAP	ΔF1 (pp)
BCE-F	0.5528	0.5900	0.5696	0.7959	0.7055	0.6319	0.5236	0.5708	0.7122	-3.0
Focal-F	0.5358	0.5816	0.6009	0.7767	0.6929	0.7937	0.5134	0.5728	0.6663	-2.8
ASL-F	0.5572	0.6334	0.6016	0.7742	0.7018	0.6736	**0.5987**	0.5974	0.6535	-0.3
BCE-G	0.5464	0.5900	0.5534	0.7959	0.7009	0.6372	0.5091	0.5633	0.7122	-3.8
Focal-G	0.5474	0.6018	0.5945	0.7767	0.7072	**0.7997**	0.5235	0.5812	0.6663	-2.0
ASL-G	0.5683	0.6254	0.6087	0.7742	0.7164	0.6392	0.5816	0.6008	0.6535	+0.0
TALC-G	**0.5737**	**0.7599**	**0.6286**	**0.8073**	**0.7383**	0.7920	0.5867	**0.6541**	**0.7387**	**+5.3**

VI. CONCLUSION

This paper studied long-tailed multi-label remote-sensing recognition from the perspective of score-to-label conversion. In this setting, rare labels may receive useful ranking scores but still be omitted from the final image-level label set after thresholding. TALC addresses this label-delivery stage by combining ASL-based score learning, group-level decision-aware training, and validation-selected H/M/T operating boundaries.

On China-MAS-50k, TALC improved tail-label delivery across multiple backbone carriers. The fixed 0.50 results first showed a stronger TALC score source, and the Agriculture-Vision-2021 evaluation showed a similar tail-label pattern in a second high-resolution agricultural imagery setting.

Together, these results show that TALC improves tail-label delivery primarily by strengthening the pre-threshold score source, while compact group-wise boundaries provide the final interface for label-set generation.

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